

Roof Lifecycle Cost Analysis

Planned maintenance is projected to save the owner of a typical Laurel office building **\$537K** over the next 20 years.

That is 57.9% less than the reactive cost, or \$230K in today's dollars. It compares PaxSeal planned maintenance with fixing problems only after they leak, using Laurel costs and storm history.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE	NEW ROOF TODAY
Laurel	Office	40,000 sq ft	TPO	8 yrs Good	\$308K
PAXSEAL NET COST		REACTIVE NET COST		VALUE PROTECTED	
\$390K		\$927K		\$186K	
\$731K spent, less \$341K roof life left		\$977K spent, less \$50K roof life left		\$14K/yr more NOI at a 7.5% cap rate	
				NEW ROOF PRICE	
				2× in 15 yrs	
				\$308K today, \$757K by year 20	

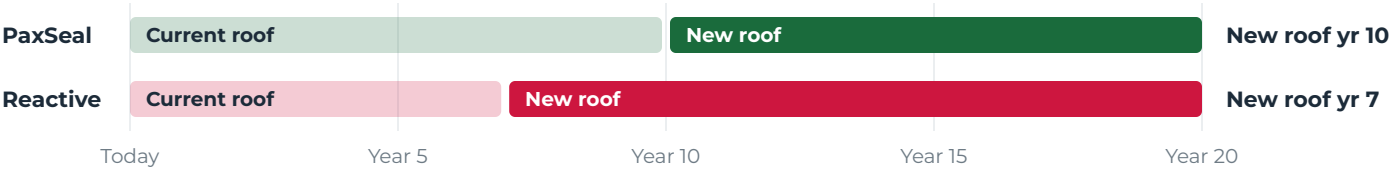
Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal program fees	\$161K	—	-\$161K
Repairs paid by owner	\$74K	\$536K	+\$462K
Expected storm damage	\$13K	\$19K	+\$6K
Roof replacements	\$483K	\$422K	-\$61K
Credit for roof life left	-\$341K	-\$50K	+\$290K
Net 20-year cost	\$390K	\$927K	+\$537K

What this means

- 1 A later replacement. Maintenance moves the next new roof from year 7 to year 10, so fewer roofs are bought over the period.
- 2 Smaller repairs. Problems found on a scheduled visit avoid emergency work, which costs about 3× as much. That saves \$462K in repairs.
- 3 The program pays for itself. Each \$1 of PaxSeal fees returns about \$3.33 in avoided cost.

Local risk profile

Prince George's County, MD (24033)

Thunderstorm wind / derecho, snow load

~3.5 major regional roof-damaging events per 20 years (2010 snow, 2012 derecho, 2016 blizzard). Expected storm damage to this roof over 20 years is **\$13K** if maintained and **\$19K** if neglected. A documented maintenance history also helps when an insurer questions a claim.

Hail	Moderate	■■■■■
Strong wind	High	■■■■■
Tornado	High	■■■■■
Hurricane	Moderate	■■■■■
Winter weather	Very High	■■■■■
Ice storm	Moderate	■■■■■

FEMA National Risk Index rating

Key assumptions

Confidence reflects source strength. Values will be refined as research continues.

ASSUMPTION	VALUE	BASIS	CONFIDENCE
Roof life	20 / 15 yrs	Maintained / reactive. Fannie Mae service-life tables; +5 yrs from NRC Canada roof study and manufacturer warranty terms	MEDIUM
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing contractors, 2007–2025	HIGH
Replacement cost	\$7.70/sq ft	Laurel contractor price ranges	LOW
Reactive repair cost	3× planned	Industry estimates; no primary study found	LOW
Storm damage	0.13%/yr	Of roof cost. FEMA NRI expected annual building loss from hail, wind, tornado, hurricane and winter weather	MEDIUM
PaxSeal fee	\$6,000/yr	PAX program estimate for this roof size. Your proposal will state the actual fee	ESTIMATE
Leak consequences	1.20× warehouse	Scales reactive repair costs. Order supported by school-closure, hospital infection-control and insurer water-damage evidence; exact values are estimates	LOW
Cap rate	7.5%	Laurel office market reports	LOW

Estimates, not a quote. Projections use Laurel market research and industry data, in nominal dollars; actual costs will differ. The PaxSeal fee is an estimate and your proposal will state the actual price. Net cost credits the roof life remaining at year 20. Property value is the yearly gain in net operating income divided by the cap rate, and applies to income-producing property where the owner pays for the roof.